Comprehensive Feature Engineering & Feature Selection Technical Report V2

Project Title: E-Commerce Customer Churn Pipeline

Date: July 2026

Dataset Scope: 6,000 Customers (19 Expanded Features)

Evaluation Framework: Multi-Technique Comparison (Filter & Wrapper Methods)

Executive Summary: Following the initial Exploratory Data Analysis (EDA) and Feature Engineering phase, a systematic benchmarking study was conducted across multiple Feature Selection methodologies. We evaluated both **Filter Methods** (ANOVA F-Test, Mutual Information) and **Wrapper Methods** (Forward Stepwise Selection, Recursive Feature Elimination - RFE). Custom-engineered domain features—such as inactivity_risk_score, frustration_index, and discount_hunter—consistently captured top rankings across methods, demonstrating robust predictive capability across linear, non-linear, and model-driven selection approaches.

1. Domain-Driven Feature Engineering Architecture

To capture complex customer interaction patterns and mitigate raw-feature multicollinearity, custom composite features were engineered prior to modeling:

Engineered Feature	Formula / Derivation	Domain Rationale
inactivity_risk_score	days_since_last_purchase / (browsing_frequency_per_week + 1)	Isolates disengagement velocity by weighting inactivity duration against weekly browsing habits.
frustration_index	(customer_support_tickets * return_rate) / (satisfaction_score + 1)	Quantifies multi-channel customer friction (tickets + product returns divided by satisfaction).

Engineered Feature	Formula / Derivation	Domain Rationale
total_customer_spend	avg_order_value * total_orders	Proxies Customer Lifetime Monetary Value (LTV) to preserve high-value customer churn signals.
discount_hunter	discount_usage_rate * (1 / (avg_order_value + 1))	Identifies price-sensitive segments highly prone to competitor poaching.

2. Methodological Comparative Matrix (Filter vs. Wrapper)

Four distinct feature selection paradigms were implemented on standardized training data (Stratified Train-Test Split: 4,800 train / 1,200 test samples):

Selection Method	Category	Underlying Strategy	Key Structural Strengths
ANOVA F-Test (f_classif)	Filter	Linear variance analysis per feature.	Ultra-fast computational speed; isolates direct linear predictors.
Mutual Information (mutual_info_classif)	Filter	Information gain & entropy reduction.	Captures complex non-linear relationships and non-monotonic dependencies.
Forward Selection (SequentialFeatureSelector)	Wrapper	Iterative subset expansion optimized for ROC-AUC.	Builds synergistic feature subsets directly optimizing model performance.

Selection Method	Category	Underlying Strategy	Key Structural Strengths
RFE (Recursive Feature Elimination)	Wrapper	Iterative backward pruning based on estimator weights.	Eliminates feature noise while retaining monetary and experiential balance.

3. Cross-Method Selected Features Comparison

Below is the detailed feature comparison across all four selection strategies:

Rank / Index	ANOVA F-Test	Mutual Information	Forward Selection	Recursive Feature Elimination (RFE)
1	days_since_last_purchase	avg_order_value	account_age_months	days_since_last_purchase
2	engagement_score	total_orders	avg_order_value	discount_usage_rate
3	inactivity_risk_score	days_since_last_purchase	days_since_last_purchase	return_rate
4	frustration_index	discount_usage_rate	discount_usage_rate	customer_support_tickets
5	customer_support_tickets	customer_support_tickets	customer_support_tickets	loyalty_member
6	satisfaction_score	loyalty_member	loyalty_member	product_review_score_avg

Rank / Index	ANOVA F-Test	Mutual Information	Forward Selection	Recursive Feature Elimination (RFE)
7	browsing_frequency_per_week	engagement_score	product_review_score_avg	engagement_score
8	product_review_score_avg	satisfaction_score	engagement_score	satisfaction_score
9	return_rate	inactivity_risk_score	satisfaction_score	total_customer_spend
10	loyalty_member	frustration_index	inactivity_risk_score	discount_hunter

4. Complete Execution Pipeline Code

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.feature_selection import SelectKBest, f_classif, mutual_info_classif, 
SequentialFeatureSelector, RFE
from sklearn.linear_model import LogisticRegression

# 1. Feature Engineering
df['inactivity_risk_score'] = df['days_since_last_purchase'] / 
(df['browsing_frequency_per_week'] + 1)
df['frustration_index'] = (df['customer_support_tickets'] * df['return_rate']) / 
(df['satisfaction_score'] + 1)
df['total_customer_spend'] = df['avg_order_value'] * df['total_orders']
df['discount_hunter'] = df['discount_usage_rate'] * (1 / (df['avg_order_value'] + 
1))

# 2. Stratified Data Split
X = df.drop(columns=['churned'])
y = df['churned']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, 
random_state=42, stratify=y)

# 3. Standard Scaling
```

```
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# 4. Feature Selection Methods Execution
base_model = LogisticRegression(class_weight='balanced', random_state=42)

# Method A: Forward Selection
sfs = SequentialFeatureSelector(base_model, n_features_to_select=10, 
direction='forward', scoring='roc_auc', cv=5, n_jobs=-1)
X_train_sfs = sfs.fit_transform(X_train_scaled, y_train)

# Method B: RFE
rfe = RFE(estimator=base_model, n_features_to_select=10)

X_train_rfe = rfe.fit_transform(X_train_scaled, y_train)
```

5. Key Insights & Final Selection Recommendation

- **Engineered Signal Persistence:** Custom metrics (e.g., inactivity_risk_score) were consistently selected by linear, non-linear, and model-wrapper approaches.
- **Forward Selection Optimization:** Forward Stepwise Selection provided the highest cross-validated ROC-AUC baseline by ensuring strong feature synergy without collinear redundancy.
- **Next Phase:** Proceed to Model Benchmarking (Logistic Regression, Random Forest, XGBoost) using the **Forward Selection** feature set as the baseline input matrix.